Statistical methods for computer science

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Introduction to Random Variables (Part II)

Where were we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

- ✓ Design phase
- ✓ Description phase
- ✓ Probability

Inference:

- Introduction to random variables
  - ✓ Discrete random variables
  - ✓ Continuous random variables

Today Multiple random variables

Course Outline

- Basic concepts
- Descriptive statistics (+ R Lab)
- Introduction to probability
- **Introduction to random variables**
- Discrete probability distributions
- Continuous probability disctributions (+ R Lab)
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

Motivation

We frequently encounter situations with multiple random variables that are interconnected and it's essential to analyze their collective behavior

Temperature and pressure in quality control

Daily returns of two different stocks

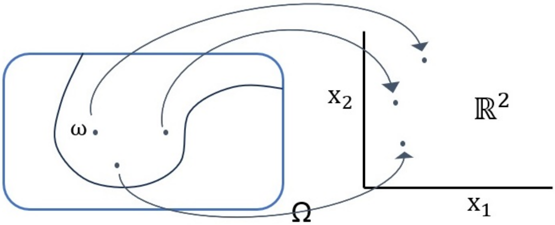
Blood pressure, cholesterol levels and heart rate

Bivariate Random Variable

Given a probability space $(\Omega, \mathcal{A}, P)$, *a bivariate random variable is a measurable function from the sample space to points in the plane* $(X_1, X_2) : \Omega \longrightarrow \mathbb{R}^2$ *such that*

$$\forall (x_1, x_2) \in \mathbb{R}^2, \{\omega : X_1(\omega) \leq x_1, X_2(\omega) \leq x_2\} \in \mathcal{A} \tag{1}$$

The measurability condition (1) is a technical condition needed to transfer probability, defined on subsets of Ω, to subsets of $\mathbb{R}^2$, i.e., $P(X_1 \leq x_1, X_2 \leq x_2) = P(\omega : X_1(\omega) \leq x_1, X_2(\omega) \leq x_2)$



set of the ω's s.t $(X_1(\omega) \leq x_1, X_2(\omega) \leq x_2)$ By assumption it belongs to $\mathcal{A}$, so it has a probability

Discrete Bivariate RV

Joint Probability Mass Function

Let (X_1, X_2) be a discrete bivariate random vector. The joint PMF of (X_1, X_2) is a function $f_{X_1,X_2} : \mathbb{R}^2 \longrightarrow \mathbb{R}$ that assigns probabilities to each possible realization of (X_1, X_2)

$$f_{X_1,X_2}(x_1, x_2) = P(X_1 = x_1, X_2 = x_2) \qquad \forall x_1, x_2 \in \mathbb{R}^2$$

The joint PMF of (X_1, X_2) completely defines the probability distribution of the random vector

Properties:

- $f_{X_1,X_2}(x_1, x_2) \geq 0$
- $\sum_{x_1,x_2 \in \mathcal{D}_f} f_{X_1,X_2}(x_1, x_2) = 1$

Example

Let's consider the random experiment "toss 2 three-sided fair dice" and define the random variables X_1 as the sum of the values appearing on the dice and X_2 as the absolute value of their difference. Find the joint PMF.

d_1	d_2	$x_1 = d_1 + d_2$	$x_2 = \|d_1 - d_2\|$
1	1	2	0
2	1	3	1
3	1	4	2
1	2	3	1
2	2	4	0
3	2	5	1
1	3	4	2
2	3	5	1
3	3	6	0

x_1, x_2	0	1	2	
2	$\frac{1}{9}$	0	0	
3	0	$\frac{2}{9}$	0	
4	$\frac{1}{9}$	0	$\frac{2}{9}$	
5	0	$\frac{2}{9}$	0	
6	$\frac{1}{9}$	0	0	
				1

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2	1	3	1
3	1	4	2
1	2	3	1
2	2	4	0
3	2	5	1
1	3	4	2
2	3	5	1
3	3	6	0

x_1, x_2	0	1	2	
2	$\frac{1}{9}$	0	0	
3	0	$\frac{2}{9}$	0	
4	$\frac{1}{9}$	0	$\frac{2}{9}$	
5	0	$\frac{2}{9}$	0	
6	$\frac{1}{9}$	0	0	
	$\frac{3}{9}$	$\frac{4}{9}$	$\frac{2}{9}$	1

How much is $P(X_2 = 0)$?

Example

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3	1	4	2
1	2	3	1
2	2	4	0
3	2	5	1
1	3	4	2
2	3	5	1
3	3	6	0

x_1, x_2	0	1	2	
2	$\frac{1}{9}$	0	0	$\frac{1}{9}$
3	0	$\frac{2}{9}$	0	$\frac{2}{9}$
4	$\frac{1}{9}$	0	$\frac{2}{9}$	$\frac{3}{9}$
5	0	$\frac{2}{9}$	0	$\frac{2}{9}$
6	$\frac{1}{9}$	0	0	$\frac{1}{9}$
	$\frac{3}{9}$	$\frac{4}{9}$	$\frac{2}{9}$	1

How much is $P(X_2 = 0)$? How much is $P(X_1 = 4)$?

Marginal Probability Mass Function

Let (X_1, X_2) be a discrete bivariate random vector with joint PMF $f_{X_1,X_2}(x_1, x_2)$. The marginal PMFs of X_1 and X_2, $f_{X_1}(x_1) = P(X_1 = x_1)$ and $f_{X_2}(x_2) = P(X_2 = x_2)$ are given by

$$f_{X_1}(x_1) = \sum_{x_2 \in \mathbb{R}} f_{X_1,X_2}(x_1, x_2) \quad f_{X_2}(x_2) = \sum_{x_1 \in \mathbb{R}} f_{X_1,X_2}(x_1, x_2)$$

- The marginals do not completely describe the joint PMF (there are many different joint distributions that have the same marginals)
- It is hopeless to try to determine $f_{X_1,X_2}(x_1, x_2)$ from knowledge of $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ only

Example

For two different cities we collected information on the variables $X_1 = \{Low, Medium, High, Very\ High\}$ and $X_2 = \{0, 1\}$ describing, respectively, cholesterol levels and dying from a stroke ($1 =$ dying).

A	L	M	H	VH	
0	92	168	444	260	964
1	0	31	275	260	566
	92	199	719	520	1530

In city A people with a very high level of cholesterol have a 50% chance of dying from a stroke, while in city B the probability of surviving is higher. Conversely, in city B there is a higher chance of dying for people with medium cholesterol levels.

B	L	M	H	VH	
0	55	44	331	265	695
1	12	99	188	110	408
	67	143	519	375	1104

Example

For two different cities we collected information on the variables $X_1 = \{Low, Medium, High, Very\ High\}$ and $X_2 = \{0, 1\}$ describing, respectively, cholesterol levels and dying from a stroke ($1 =$ dying).

A	L	M	H	VH	
0	0.06	0.11	0.29	0.17	
1	0.00	0.02	0.18	0.17	
					1

In city A people with a very high level of cholesterol have a 50% chance of dying from a stroke, while in city B the probability of surviving is higher. Conversely, in city B there is a higher chance of dying for people with medium cholesterol levels. Different joint PMFs, same marginals

B	L	M	H	VH	
0	0.05	0.04	0.30	0.24	
1	0.01	0.09	0.17	0.10	
					1

Example

For two different cities we collected information on the variables $X_1 = \{Low, Medium, High, Very\ High\}$ and $X_2 = \{0, 1\}$ describing, respectively, cholesterol levels and dying from a stroke ($1 =$ dying).

A	L	M	H	VH	
0	0.06	0.11	0.29	0.17	0.63
1	0.00	0.02	0.18	0.17	0.37
	0.06	0.13	0.47	0.34	1

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0	0.05	0.04	0.30	0.24	0.63
1	0.01	0.09	0.17	0.10	0.37
	0.06	0.13	0.47	0.34	1

If we can't recover $f_{X_1,X_2}(x_1, x_2)$ starting from $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$, what is their use?

- Sometimes we are more interested in a specific variable, e.g., we might be more interested in knowing $P(X_2 = 1)$ than $P(X_2 = 1, X_1 = H)$
- We can use them to calculate conditional probabilities, e.g., given that a person living in city B has a very high cholesterol level, what is his/her probability of surviving a stroke?

Conditional Probability Mass Function

Knowledge about the value of one variable may gives us some information about the value of another variable. This is when conditional probabilities come into play.

Let (X_1, X_2) be a discrete bivariate random vector with joint PMF $f_{X_1,X_2}(x_1, x_2)$ and marginals $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$. For any x_1 such that $f_{X_1}(x_1) > 0$, the conditional PMF of X_2 given that $X_1 = x_1$ is the function

$$f_{X_2|X_1}(x_2|x_1) = P(X_2 = x_2|X_1 = x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)}.$$

For any x_2 such that $f_{X_2}(x_2) > 0$, the conditional PMF of X_1 given that $X_2 = x_2$ is the function

$$f_{X_1|X_2}(x_1|x_2) = P(X_1 = x_1|X_2 = x_2) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_2}(x_2)}.$$

Example

Continuing previous example, let's find the probability of surviving a stroke for people with a medium and high cholesterol level living in city A and B.

1. Given medium level

$$P_A(X_2 = 0|X_1 = M) = \frac{P_A(X_2 = 0, X_1 = M)}{P_A(X_1 = M)} = \frac{0.11}{0.13} = 0.87$$

$$P_B(X_2 = 0|X_1 = M) = \frac{P_B(X_2 = 0, X_1 = M)}{P_B(X_1 = M)} = \frac{0.04}{0.13} = 0.31$$

2. Given very high level

$$P_A(X_2 = 0|X_1 = VH) = \frac{P_A(X_2 = 0, X_1 = VH)}{P_A(X_1 = VH)} = \frac{0.17}{0.34} = 0.50$$

$$P_B(X_2 = 0|X_1 = VH) = \frac{P_B(X_2 = 0, X_1 = VH)}{P_B(X_1 = VH)} = \frac{0.24}{0.34} = 0.71$$

Example

Similarly, we can compute the probability of having a high level of cholesterol for people living in both cities given that they died from a stroke.

$$P_A(X_1 = H|X_2 = 1) = \frac{P_A(X_1 = H, X_2 = 1)}{P_A(X_2 = 1)} = \frac{0.18}{0.37} = 49\%$$

$$P_B(X_1 = H|X_2 = 1) = \frac{P_B(X_1 = H, X_2 = 1)}{P_B(X_2 = 1)} = \frac{0.17}{0.37} = 46\%$$

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This can also be seen directly from the tables above after dividing for the row and column totals, e.g. for city A we have,

A	L	M	H	VH	
0	0.10	0.17	0.46	0.27	1.00
1	0.00	0.05	0.49	0.46	1.00
	0.06	0.13	0.47	0.34	1.00

Example

Similarly, we can compute the probability of having a high level of cholesterol for people living in both cities given that they died from a stroke.

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This can also be seen directly from the tables above after dividing for the row and column totals, e.g. for city A we have,

A	L	M	H	VH	
0	1.00	0.84	0.62	0.50	0.63
1	0	0.16	0.38	0.50	0.37
	1.00	1.00	1.00	1.00	1.00

Continuous Bivariate RV

Joint Probability Density Function

Let (X_1, X_2) be a continuous bivariate random vector. The joint PDF of (X_1, X_2) is a function $f_{X_1,X_2} : \mathbb{R}^2 \longrightarrow \mathbb{R}$ if for every $A \in \mathbb{R}^2$

$$P((X_1, X_2) \in A) = \int_A \int f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$$

The notation $\int_A \int$ simply means that the function is integrated over all $(x_1, x_2) \in A$

Properties:

- $f_{X_1,X_2}(x_1, x_2) \geq 0 \quad \forall x_1, x_2 \in \mathbb{R}^2$
- $\int\limits_{-\infty}^{\infty} \int\limits_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2) = 1$ (if we choose $A = \mathbb{R}$ the probability of $(X_1, X_2) \in A$ must be 1)

Marginal Probability Density Function

Let (X_1, X_2) be a continuous bivariate random vector with joint PDF $f_{X_1,X_2}(x_1, x_2)$. The marginal PDFs of X_1 and X_2, are given by

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)dx_2 \quad f_{X_2}(x_2) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)dx_1$$

- The marginals do not completely describe the joint PDF (there are many different joint distributions that have the same marginals)
- It is hopeless to try to determine $f_{X_1,X_2}(x_1, x_2)$ from knowledge of $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$ only

Conditional Probability Density Function

Let (X_1, X_2) be a continuous bivariate random vector with joint PDF $f_{X_1,X_2}(x_1, x_2)$ and marginals $f_{X_1}(x_1)$ and $f_{X_2}(x_2)$. For any x_1 such that $f_{X_1}(x_1) > 0$, the conditional PDF of X_2 given that $X_1 = x_1$ is defined by

$$f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_1}(x_1)}.$$

For any x_2 such that $f_{X_2}(x_2) > 0$, the conditional PDF of X_1 given that $X_2 = x_2$ is defined by

$$f_{X_1|X_2}(x_1|x_2) = \frac{f_{X_1,X_2}(x_1, x_2)}{f_{X_2}(x_2)}.$$

Example

Define a joint PDF as

$$f_{X,Y}(x,y) = \begin{cases} 6xy^2 & 0 < x < 1\ ,\ 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Check that $f_{X,Y}(x,y)$ is indeed a joint PDF
2. Calculate $P(X + Y \geq 1)$
3. Calculate $P(\frac{1}{2} < X < \frac{3}{4})$
4. Find the marginal $f_Y(y)$

Example

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3. Calculate $P(\frac{1}{2} < X < \frac{3}{4})$
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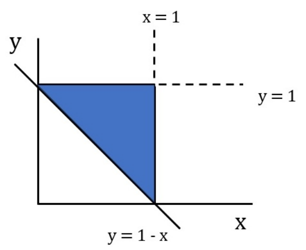
1. We must check that the joint integrates to 1

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{X,Y}(x,y)dxdy = \int_0^1\int_0^1 6xy^2 dxdy = \int_0^1 6y^2\left(\int_0^1 xdx\right)dy$$
$$= \int_0^1 3y^2 dy = 3\left[\frac{y^3}{3}\right]_0^1 = 1$$

Example

2. Define $A = \{(x, y) : x + y \geq 1\}$, to calculate the probability we need to integrate over the set A but the joint is always 0 except on the unit square, so we must integrate over the part of A that belongs to the unit square



$$P(X + Y \geq 1) = \int_{0}^{1} \int_{1-y}^{1} 6xy^2 dxdy$$

$$= \int_0^1 6y^2 \left[\int_{1-y}^{1} xdx \right] dy$$

$$= \int_0^1 6y^2 \frac{1 - (1 - y)^2}{2} dy$$

$$= \int_0^1 (6y^3 - 3y^4) dy = \frac{9}{10}$$

Example

3. The marginal $f_X(x)$ is

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dy = \int_0^1 6xy^2 dy = 6x \int_0^1 y^2 dy = 2x$$

so we can find

$$P\left(\frac{1}{2} < X < \frac{3}{4}\right) = \int_{1/2}^{3/4} 2xdx = \left[x^2\right]_{1/2}^{3/4} = \frac{5}{16}$$

4. The marginal $f_Y(y)$ is

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dx = \int_0^1 6xy^2 dx = 6y^2 \int_0^1 xdx = 3y^2$$

What can we notice about the marginals and the joint?

Joint CDF

As in the univariate case, we can also define the cumulative distribution function but since we have a bivariate vector, as with the PMF/PDF, we can define the joint, marginal and conditional distribution functions

Let (X_1, X_2) be a bivariate random vector. The joint CDF of (X_1, X_2) is a function $F_{X_1,X_2} : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined as

$$F_{X_1,X_2}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2) \qquad \forall x_1, x_2 \in \mathbb{R}^2$$

Link between joint CDF and joint PDF/PMF

As in the univariate case, there is a direct relationship between the joint PMF or PDF and the joint CDF:

- For a discrete bivariate random vector

$$P(X_1 \leq x_1, X_2 \leq x_2) = \sum_{x_i \leq x_1} \sum_{x_j \leq x_2} P(X_1 = x_i, X_2 = x_j)$$

$$P(X_1 = x_i, X_2 = x_j) = P(X_1 \leq x_i, X_2 \leq x_j) - P(X_1 \leq x_{i-1}, X_2 \leq x_{j-1})$$

- For a continuous bivariate random vector

$$F_{X_1,X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1,X_2}(s, t) dtds$$

$$f_{X_1,X_2}(x_1, x_2) = \frac{\partial^2 F_{X_1,X_2}(x_1, x_2)}{\partial x_1 \partial x_2}$$

Marginal and Conditional CDFs

Starting from a bivariate discrete or continuous random vector (X_1, X_2), *the marginal CDFs are given, respectively, by*

$$F_{X_1}(x_1) = \sum_{x_2} F_{X_1,X_2}(x_1, x_2) \ , \ F_{X_1}(x_1) = \int_{-\infty}^{+\infty} F_{X_1,X_2}(x_1, x_2) dx_2$$

$$F_{X_2}(x_2) = \sum_{x_1} F_{X_1,X_2}(x_1, x_2) \ , \ F_{X_2}(x_2) = \int_{-\infty}^{+\infty} F_{X_1,X_2}(x_1, x_2) dx_1$$

For any x_2 *such that* $F_{X_2}(x_2) > 0$, *the conditional CDF for* X_1 *given* X_2 *is*

$$F_{X_1|X_2}(x_1|x_2) = \frac{F_{X_1,X_2}(x_1, x_2)}{F_{X_2}(x_2)} \ , \ F_{X_1|X_2}(x_1|x_2) = \frac{\int\limits_{-\infty}^{x_1} \int\limits_{-\infty}^{x_2} F_{X_1,X_2}(s, t) ds dt}{\int\limits_{-\infty}^{+\infty} F_{X_1,X_2}(x_1, x_2) dx_2}$$

Example

1. In the dice experiment, find $P(X_1 \leq 5, X_2 \leq 1)$ (recall that X_1 is the sum of the values appearing on the dice and X_2 is their difference)

d_1	d_2	$x_1 = d_1 + d_2$	$x_2 = \lvert d_1 - d_2 \rvert$
1	1	2	0
2	1	3	1
3	1	4	2
1	2	3	1
2	2	4	0
3	2	5	1
1	3	4	2
2	3	5	1
3	3	6	0

x_1, x_2	0	1	2	
2	$\frac{1}{9}$	0	0	$\frac{1}{9}$
3	0	$\frac{2}{9}$	0	$\frac{2}{9}$
4	$\frac{1}{9}$	0	$\frac{2}{9}$	$\frac{3}{9}$
5	0	$\frac{2}{9}$	0	$\frac{2}{9}$
6	$\frac{1}{9}$	0	0	$\frac{1}{9}$
	$\frac{3}{9}$	$\frac{4}{9}$	$\frac{2}{9}$	1

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d_1	d_2	$x_1 = d_1 + d_2$	$x_2 = \lvert d_1 - d_2 \rvert$
1	1	2	0
2	1	3	1
3	1	4	2
1	2	3	1
2	2	4	0
3	2	5	1
1	3	4	2
2	3	5	1
3	3	6	0

x_1, x_2	0	1	2	
2	$\frac{1}{9}$	0	0	$\frac{1}{9}$
3	0	$\frac{2}{9}$	0	$\frac{2}{9}$
4	$\frac{1}{9}$	0	$\frac{2}{9}$	$\frac{3}{9}$
5	0	$\frac{2}{9}$	0	$\frac{2}{9}$
6	$\frac{1}{9}$	0	0	$\frac{1}{9}$
	$\frac{3}{9}$	$\frac{4}{9}$	$\frac{2}{9}$	1

$$P(X_1 \leq 5, X_2 \leq 1) = P(X_1 = 2, X_2 = 0) + P(X_1 = 3, X_2 = 1) + P(X_1 = 4, X_2 = 0)$$
$$+ P(X_1 = 5, X_2 = 1) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} + \frac{2}{9} = \frac{6}{9}$$

Example

2. Continuing the above example where $f_{X,Y}(x,y) = 6xy^2$ for $0 < x < 1$ and $0 < y < 1$, calculate $P\left(Y < \frac{1}{4} | \frac{1}{2} < X < \frac{3}{4}\right)$ and show that it is equal to $P\left(Y < \frac{1}{4}\right)$

Example

2. Continuing the above example where $f_{X,Y}(x,y) = 6xy^2$ for $0 < x < 1$ and $0 < y < 1$, calculate $P\left(Y < \frac{1}{4} | \frac{1}{2} < X < \frac{3}{4}\right)$ and show that it is equal to $P\left(Y < \frac{1}{4}\right)$

$$P\left(Y < \frac{1}{4} \Big| \frac{1}{2} < X < \frac{3}{4}\right) = \frac{P\left(Y < \frac{1}{4}, \frac{1}{2} < X < \frac{3}{4}\right)}{P\left(\frac{1}{2} < X < \frac{3}{4}\right)}$$

$$= \frac{\int_0^{1/4}\int_{1/2}^{3/4} 6xy^2 dxdy}{\int_{1/2}^{3/4} 2xdx} = \frac{16}{5}\int_0^{1/4}\int_{1/2}^{3/4} 6xy^2 dxdy$$

$$= \frac{16}{5}\int_0^{1/4} 6y^2 \left[\frac{x^2}{2}\right]_{1/2}^{3/4} dy = 3\int_0^{1/4} y^2 dy = 3\left[\frac{y^3}{3}\right]_0^{1/4} = \frac{1}{64}$$

$$P\left(Y < \frac{1}{4}\right) = \int_0^{1/4} 3y^2 dy = 3\left[\frac{y^3}{3}\right]_0^{1/4} = \frac{1}{64}$$

Summary

- Discrete bivariate random vector $\longrightarrow$ PMF
  - Joint PMF
  - Marginal PMF
  - Conditional PMF
- Continuous bivariate random vector $\longrightarrow$ PDF
  - Joint PDF
  - Marginal PDF
  - Conditional PDF
- CDF
  - Joint CDF
  - Marginal CDF
  - Conditional CDF

Expectations

Expectations

Let X_1, X_2 be a discrete random vector whose joint PMF is given by $f_{X_1,X_2}(x_1, x_2)$ and let $g(x_1, x_2)$ real valued function defined for all possible values (x_1, x_2) of the discrete random vector. Then $g(X_1, X_2)$ is itself a random variable and its expected value is given by

$$E[g(X_1, X_2)] = \sum_{x_1, x_2 \in \mathbb{R}^2} g(X_1, X_2) f_{X_1,X_2}(x_1, x_2)$$

Let X_1, X_2 be a continuous random vector whose joint PDF is given by $f_{X_1,X_2}(x_1, x_2)$ and let $g(x_1, x_2)$ real valued function defined for all possible values (x_1, x_2) of the continuous random vector. Then $g(X_1, X_2)$ is itself a random variable and its expected value is given by

$$E[g(X_1, X_2)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(X_1, X_2) f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$$

Expectations

Some expectations have been already considered, at least implicitly. For example, from the joint PMF we can compute,

$$\mu_1 = E[X_1] = \sum_{x_1} x_1 f_{X_1}(x_1) = \sum_{x_1} x_1 \sum_{x_2} f_{X_1,X_2}(x_1, x_2)$$

$$= \sum_{x_1} \sum_{x_2} x_1 f_{X_1,X_2}(x_1, x_2)$$

$$\sigma_1^2 = VAR[X_1] = \sum_{x_1} (x_1 - \mu_1)^2 f_{X_1}(x_1) = \sum_{x_1} (x_1 - \mu_1)^2 \sum_{x_2} f_{X_1,X_2}(x_1, x_2)$$

$$= \sum_{x_1} \sum_{x_2} (x_1 - \mu_1)^2 f_{X_1,X_2}(x_1, x_2)$$

and the same can be done for X_2.

Expectations

Similarly, from the joint PDF we can compute,

$$\mu_1 = E[X_1] = \int_{-\infty}^{+\infty} x_1 f_{X_1}(x_1) dx_1 = \int_{-\infty}^{+\infty} x_1 \left(\int_{-\infty}^{+\infty} f_{X_1,X_2}(x_1, x_2) dx2 \right) dx_1$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_1 f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$$

$$\sigma_1^2 = VAR[X_1] = \int_{-\infty}^{+\infty} (x_1 - \mu_1)^2 f_{X_1}(x_1) dx_1 = \int_{-\infty}^{+\infty} (x_1 - \mu_1)^2 \left(\int_{-\infty}^{+\infty} f_{X_1,X_2}(x_1, x_2) dx_2 \right) dx_1$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x_1 - \mu_1)^2 f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$$

and same can be done for X_2.

Expectations

We can now compute expectations involving both components, e.g., we can compute the mean of the product and the covariance

$$
\begin{aligned}
E[X_1X_2] &= \sum_{x_1}\sum_{x_2}(x_1x_2)f_{X_1,X_2}(x_1,x_2)\\
COV[X_1,X_2] &= \sigma_{12} = E[(X_1-\mu_1)(X_2-\mu_2)]\\
&= \sum_{x_1}\sum_{x_2}(x_1-\mu_1)(x_2-\mu_2)f_{X_1,X_2}(x_1,x_2)
\end{aligned}
$$

$$
\begin{aligned}
E[X_1X_2] &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(x_1x_2)f_{X_1,X_2}(x_1,x_2)dx_1dx_2\\
COV[X_1,X_2] &= \sigma_{12} = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}(x_1-\mu_1)(x_2-\mu_2)f_{X_1,X_2}(x_1,x_2)dx_1dx_2
\end{aligned}
$$

Expectation

Note that we can use the following shortcut for covariance computation,

$$COV[X_1, X_2] = E[X_1 X_2] - E[X_1]E[X_2]$$

Proof:

$$\begin{aligned}
COV[X_1, X_2] &= E[(X_1 - \mu_1)(X_2 - \mu_2)] = E[X_1 X_2 - X_1\mu_2 - X_2\mu_1 + \mu_1\mu_2] \\
&= E[X_1 X_2] - E[X_1]E[X_2] - E[X_2]E[X_1] + E[X_1]E[X_2] \\
&= E[X_1 X_2] - 2E[X_1]E[X_2] + E[X_1]E[X_2] \\
&= E[X_1 X_2] - E[X_1]E[X_2]
\end{aligned}$$

Expectations

Some properties:

1. $COV[X_1, X_1] = VAR[X_1]$
2. $COV[X_1, X_2] = COV[X_2, X_1]$

For any constants $a, b, c, d \in \mathbb{R}$ it holds

3. $COV[X_1, a] = 0$
4. $COV[aX_1, bX_2] = abCOV[X_1, X_2]$
5. $COV[X_1 + a, X_2 + b] = COV[X_1, X_2]$
6. $COV[aX_1 + bY_1, cX_2 + dY_2] =$ $acCOV[X_1, X_2] + adCOV[X_1, Y_2] + bcCOV[Y_1, X_2], bdCOV[Y_1, Y_2]$

The proof is left as an exercise

Expectations

Another important expectation is the linear correlation coefficient, defined for both discrete and continuous bivariate random vectors as,

$$\rho_{12} = \frac{COV[X_1, X_2]}{\sqrt{VAR[X_1]}\sqrt{VAR[X_2]}} = \frac{\sigma_{12}}{\sigma_1 \sigma_2}$$

Remarks:

- Covariance and correlation measure the linear relationship between X_1 and X_2
- Correlation is bounded between -1 (perfect negative linear relation) and 1 (perfect positive linear relation)
- If X_1 and X_2 are independent then they are uncorrelated but the reverse is not true (the only exception is when X_1, X_2 are jointly Normal)

Expectations

Where do the bounds $-1 \leq \rho_{X_1,X_2} \leq 1$ come from?

Proof:
Denote with σ_{XY} the covariance between two random variables X and Y with variances σ_Y^2, σ_X^2 and let $Z = X - \frac{\sigma_{XY}}{\sigma_Y^2} Y$. Then,

$$
\begin{aligned}
VAR[Z] &= COV\left[X - \frac{\sigma_{XY}}{\sigma_Y^2} Y, X - \frac{\sigma_{XY}}{\sigma_Y^2} Y\right] \\
&= COV[X, X] - 2\frac{\sigma_{XY}}{\sigma_Y^2} COV[X, Y] + \frac{(\sigma_{XY})^2}{\sigma_Y^2 \sigma_Y^2} COV[Y, Y] \quad \text{by prop. 6} \\
&= \sigma_X^2 - 2\frac{(\sigma_{XY})^2}{\sigma_Y^2} + \frac{(\sigma_{XY})^2}{\sigma_Y^2} \\
&= \sigma_X^2 - \frac{(\sigma_{XY})^2}{\sigma_Y^2} \geq 0 \Leftrightarrow (\sigma_{XY})^2 \leq \sigma_X^2 \sigma_Y^2 \Leftrightarrow -\sigma_X\sigma_Y \leq \sigma_{XY} \leq \sigma_X\sigma_Y
\end{aligned}
$$

Finally, by dividing all terms for the product $\sigma_X\sigma_Y$, $-1 \leq \rho_{X,Y} \leq 1$

Conditional Expectations

Conditional PMF or PDF can also be used to calculate expected values. Just remember that $f_{X_1|X_2}(x_1|x_2)$ is a PMF or PDF itself and we can use it in the same way that we used the joint PMF or PDF.

Let $g(X_1)$ a real-valued function of X_1, then the conditional expectation of $g(X_1)$ given X_2 is

$$E[g(X_1)|X_2] = \sum_{x_1} g(x_1) f_{X_1|X_2}(x_1|x_2)$$

$$E[g(X_1)|X_2] = \int_{-\infty}^{+\infty} g(x_1) f_{X_1|X_2}(x_1|x_2) dx_1$$

Conditional Expectations

By letting $g(X_1) = 1 \cdot X_1$ and $g(X_1) = (X_1 - \mu_1)^2$, we can also define the conditional expected value and the conditional variance

For a discrete random vector (X_1, X_2),

$$E[X_1|X_2] = \sum_{x_1} x_1 f_{X_1|X_2}(x_1|x_2)$$

$$E[(X_1 - \mu_1)^2|X_2] = \sum_{x_1} (x_1 - \mu_1)^2 f_{X_1|X_2}(x_1|x_2)$$

For a continuous random vector (X_1, X_2),

$$E[X_1|X_2] = \int_{-\infty}^{+\infty} x_1 f_{X_1|X_2}(x_1|x_2) dx_1$$

$$E[(X_1 - \mu_1)^2|X_2] = \int_{-\infty}^{+\infty} (x_1 - \mu_1)^2 f_{X_1|X_2}(x_1|x_2) dx_1$$

Summary

To sum up, for both discrete and continuous bivariate vectors, we can define

- "Joint" expectations $\longrightarrow$ defined from the joint PMF or PDF, meaning the expectations of any given function of both variables (X_1, X_2), e.g., expectation of the product, covariance and so on
- Marginal expectations $\longrightarrow$ after marginalizing the joint PMF or PDF
- conditional expectations $\longrightarrow$ defined from the conditional PMF or PDF

Independence

Let (X_1, X_2) be a bivariate random vector with joint PDF or PMF $f_{X_1,X_2}(x_1,x_2)$ and marginal PDFs or PMFs $f_{X_1}(x_1)$, $f_{X_2}(x_2)$. Then, X_1, X_2 are called independent random variables if, for every $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$,

$$f_{X_1,X_2}(x_1,x_2) = f_{X_1}(x_1) f_{X_2}(x_2)$$

or, alternatively

$$F_{X_1,X_2}(x_1,x_2) = F_{X_1}(x_1) F_{X_2}(x_2)$$

Remark: if X_1, X_2 are independent

- $f_{X_1|X_2}(x_1|x_2) = f_{X_1}(x_1)$, i.e. knowing x_2 gives us no additional information on x_1
- $f_{X_2|X_1}(x_2|x_1) = f_{X_2}(x_2)$,i.e. knowing x_1 gives us no additional information on x_2
- This form of independence is also known as marginal independence

Exercise

We sampled 141 firms and collected information on how many attacks to their websites they received last month and the implemented security measures. The table below shows the number of attacks (Y) as increasing securities measures are taken (X).

	Y = 2	Y = 4	Y = 5
X = 1	42	18	12
X = 2	24	16	8
X = 3	12	6	3

1. Find the probability of having more than 4 attacks and a number of security measures less or equal 2
2. Find the marginal PMFs of X and Y
3. Given that we implemented only 1 security measure, find the probability of receiving exactly 2 attacks
4. Are X and Y independent?
5. What is is the number of attacks we can expect if we implement 3 security measures?